Transfer: gravity and motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply gravity, projectile and orbital reasoning to a new set of cases.

SUCCESS CRITERIA - I can...

1. Choose a diagram or equation. State what it treats as true before calculating.
2. Connect diagrams to forces and motion.
3. Check units and explain the limits of conclusions.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A circular orbital radius becomes **9** times as large around the same source. **Compare orbital speed** and period.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_g = mg$; $a_c = v^2/r$; $h = \frac{1}{2}gt^2$; range = $v_x t$.
Match each equation to its case: near-surface weight, circular motion, or horizontal launch. Use g = **10 m/s²**.

Worked reasoning: A toy horizontal projectile falls for **2 s** while moving at **4 m/s**. Range is **8 m**. The vertical motion sets time; the horizontal motion sets range once time is known.

Your turn: Draw two coordinated timelines, one for each direction, with a shared **2 s** endpoint.

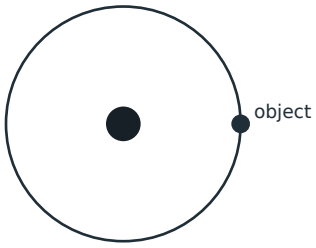


Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A satellite moves in a circular orbit.
1. **Draw velocity** and gravity arrows.
2. **Explain why floating** does not mean zero gravity.



The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball launches horizontally at **6 m/s** from **20 m**. **Find time**, range, final vertical velocity and final speed. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

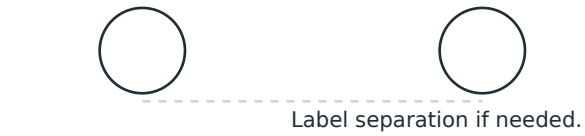
05 Explain your reasoning

A Sun-orbit returns every **2 years**. Why can period alone not prove an Earth encounter? **Name one measurement or observation** you would request.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

For fixed masses, what separation multiplier makes gravity **1/16** of its original value? What multiplier is needed if both masses also become **6** times as large? **Label the masses**, field or force relevant to the question.



Label separation if needed.

Source Object

Marzano check - circle: 1 2 3

07 Exit check

For GM=**400 m³/s²** and circular r=**25 m**, find speed, acceleration and period. **Explain what changes** in velocity. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A horizontal projectile launches from a **45 m** platform at **12 m/s**; g=**10 m/s²**. **Find flight time**, range and final velocity. Then halve g with all starting values fixed and **compare range**. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3